

Project Design Phase Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	4 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

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Solution Architecture:

The **AI WeatherWise API** is designed as a secure and scalable RESTful backend application that integrates **real-time weather services** with **Google Gemini AI** to provide intelligent weather summaries and personalized recommendations.

The architecture follows the **MVC (Model-View-Controller) pattern**, ensuring separation of concerns, maintainability, and scalability.

Example - Solution Architecture Diagram:

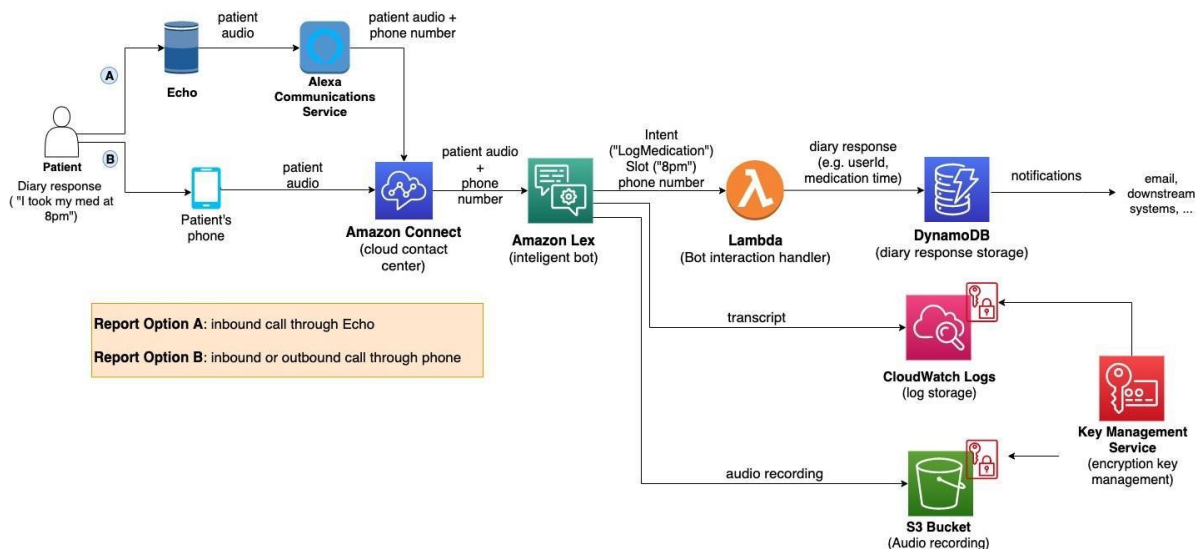


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

1. Client Layer

- Users access the system through Postman, Thunder Client, Web Application, or Mobile Application.
- Sends HTTP requests to the backend APIs.
- Receives JSON responses.

2. Express Server Layer

- Built using Node.js and Express.js.
- Handles incoming requests.
- Configures middleware, routing, CORS, and request processing.

3. Authentication Layer

- User Registration and Login.
- Password encryption using bcryptjs.
- JWT token generation and validation.
- Protects authorized routes.

4. Controller Layer

- Processes user requests.
- Handles weather retrieval operations.
- Manages favorite locations.
- Coordinates Gemini AI interactions.
- Returns structured API responses.

5. Service Integration Layer

- OpenWeather API for real-time weather information.
- Google Gemini AI for:
 - Weather summaries
 - Clothing recommendations
 - Outdoor activity suggestions
 - Safety advice

6. Database Layer

- MongoDB database with Mongoose ODM.
- Stores:
 - User information
 - Authentication details
 - Favorite locations
- Maintains relationships between users and saved locations.

```
WeatherWiseAI
├── Backend
│   ├── node_modules
│   └── src
│       ├── config
│       │   └── db.js
│       ├── controllers
│       │   ├── aiController.js
│       │   ├── authController.js
│       │   ├── locationController.js
│       │   └── weatherController.js
│       ├── middleware
│       │   └── authMiddleware.js
│       ├── models
│       │   ├── Location.js
│       │   └── User.js
│       └── routes
│           ├── aiRoutes.js
│           ├── authRoutes.js
│           ├── locationRoutes.js
│           └── weatherRoutes.js
├── services
│   ├── aiService.js
│   └── weatherService.js
├── app.js
├── server.js
├── .env
├── .gitignore
├── package-lock.json
├── package.json
├── postman_collection.json
└── README.md
```